

IPhO Style Multi Problem Paper

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·1· Mechanics warmup

A block slides on an inclined plane.

Find the acceleration and normal force.

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2. Oscillation setup

A mass-spring oscillator is released from rest.

Derive the period and maximum speed.

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Continuation of problem 2

Use energy conservation for the final expression.

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(3) Thermodynamics cycle

An ideal gas follows an isothermal and adiabatic path.

Compute the work done in one cycle.

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Problem 4 Electromagnetic induction

A conducting rod moves through a uniform magnetic field.

Determine the induced emf.

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Continuation of Problem 4

Discuss the direction of current and magnetic braking.

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·5 Geometric optics

A lens system forms an image on a screen.

Find the final image position and magnification.